



CS5071NI - Professional and Ethical Issues

100% Individual Coursework

2024-25 Spring

Credit: 15 Semester Long Module

Student Name: Carron Singh

London Met ID: 23050380

College ID: NP01CP4A230414

Assignment Due Date: Monday, May 19th, 2025,

Assignment Submission Date: Monday, May 19th, 2025

Word Count: 3500

Submitted to: Aadesh Tandukar

I confirm that I understand my coursework needs to be submitted online via MySecondTeacher under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a marks of zero will be awarded.

23050380 Carron Singh.docx

 Islington College, Nepal

Document Details

Submission ID

trn:oid:::3618:96553606

Submission Date

May 19, 2025, 1:31 AM GMT+5:45

Download Date

May 19, 2025, 1:32 AM GMT+5:45

File Name

23050380 Carron Singh.docx

File Size

39.3 KB

40 Pages

5,624 Words

32,561 Characters



Page 2 of 46 - Integrity Overview

Submission ID trn:oid:::3618:96553606

15% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **54 Not Cited or Quoted 12%**
Matches with neither in-text citation nor quotation marks
-  **12 Missing Quotations 3%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **1 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 8%  Internet sources
- 1%  Publications
- 14%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Table of Contents

1. Introduction.....	5
1.1 Introduction to Cyber Threats and Risks	6
1.2 Background of the Scandal	8
2. Social Issues.....	10
2.1 Background on Social Issues	10
2.2 Identified Social Issues	10
3. Ethical Issues	13
3.1 Background on Ethical issues	13
3.2 Identified Ethical Issues.....	13
3.3 Identified ethical issues to be compared with the general ethical theories.	16
4. Legal Issue	17
4.1 Background on Legal Issues	17
4.2 Identified legal Issues	17
4.3 Identified legal issues related with different laws, court cases, legal actions, fines and settlements and so on.	18
5.1 Background on Professional Issues	19
5.2 Identified Professional Issues	19
5.3 Identified Professional issues to be compared with the Professional Code of Conduct	20
6. Conclusion	21
6.2 Personal Reflection	22
6.3 Recommendations.....	22
6.4 References.....	24
7. Appendix.....	29
7.1. The 5 legal issues are mentioned in appendix A.....	29
7.2. Identified legal issues related with different laws, court cases, legal actions, fines and settlements and so on. are mentioned in appendix B.....	32
7.3. Identified Professional Issues are mentioned in appendix C	33

Table of Figure

Figure 1 Capital One.....	5
Figure 2 Number of individuals affected by the Capital One breach	9
Figure 3 Ethical issues to be compared with the general ethical theories	16
Figure 4 legal issues related to different laws.....	33

1. Introduction

Capital One Financial Corporation is one of the best leading American bank holding companies which offer many services like credit cards, loans , financing, and banking . This bank was started in 1994, Richard Fairbank is the founder, Chairperson, and CEO of Capital One, The Vanguard Group, BlackRock, and Dodge & Cox are the largest institutional shareholders (Team, 2023). Its head office is located in McLean, Virginia (Anon., 2019). Capital One is one of the top biggest banks in the US by total assets . Capital One is most famous for its approach in tech and early adoption in AWS and cloud computing to make and give customers a better experience.

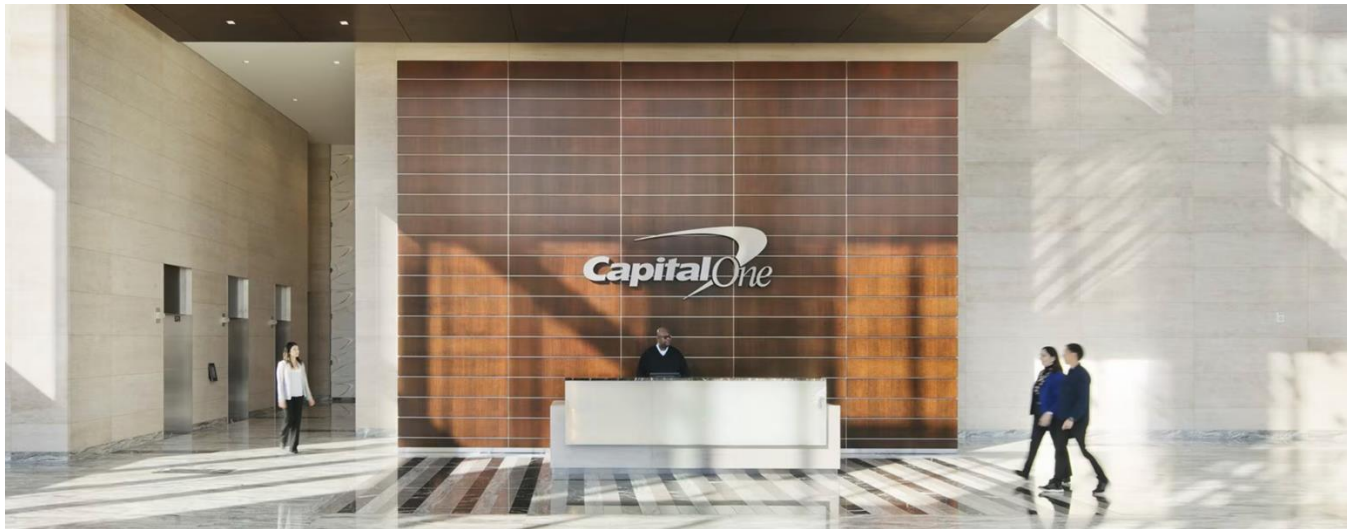


Figure 1 Capital One

Capital One tried to be a forward-thinking company but it came to be a target of modern crimes related to cyber threats. In today's world organizations are facing the risk of protecting their data during the digital era . However, today businesses are leaning more about cloud infrastructure , artificial intelligence, and big data analytics; it becomes more at risk to cyber-attacks . (Olivér Gulyás, 2023).

Organizations today must protect themselves from several cyber threats, including data breaches, ransomware attacks, denial-of-service (DoS) attacks and social engineering. (Folorunso, 2024) These threats have the potential to seriously harm an organization financially and could also damage its reputation and the confidence its consumers place in them. (Solove, 2006) points out that there is a serious

ethical problem with privacy violations nowadays, because individuals generally have no control over how their data is stored and looked after online.

1.1 Introduction to Cyber Threats and Risks

Cyber threats are now one of the biggest dangers that organizations face in the digital world all over the world. Different types of data breaches or phishing tries lead to cyber-attacks. this types of attacks including ransomware and threats come from within the company and this puts the user at risk of identity theft. (Ghelani, 2022)

Few points:

Phishing Attacks

Phishing attack has become one of the most prominent attacks faced by internet users, governments, and service-providing organizations. In a phishing attack, the attacker(s) collects the client's sensitive data (i.e., user account login details, credit/debit card numbers, etc.) by using spoofed emails or fake websites. (Zafar, 2021)

Example: A phishing attack against Twitter employees in 2020 caused employee account takeovers which resulted in a hacker invasion of the platform. (Wikipedia, 2020)

Malware and Ransomware

Software programs have been designed to both interrupt operating systems and essential data.

Example: In 2017 the WannaCry ransomware encryption targeting hospitals ,banks and businesses spread across the globe to demand payments. (Kao, 2018)

Internal Threats (Insider Attacks)

Former or present workforce members take advantage of their authorized privileges.

Example: During the Capital One breach Paige Thompson who used to work at AWS leveraged their cloud expertise to commit the attack. (Daswani, 2021)

Server-Side Request Forgery (SSRF)

The vulnerability allows malicious parties to exploit servers by issuing requests which led to the Capital One data breach.

Example: The Capital One data breach resulted from Thompson taking advantage of an SSRF flaw to penetrate the cloud-stored database. (Daswani, 2021)

Distributed Denial-of-Service (DDoS) Attacks

Hackers generate server traffic to make operations functional.

Example: During 2016 Dyn DNS attacked massive internet sites such as Twitter and Netflix. (Greenstein, 2019)

1.2 Background of the Scandal

In September 2017 Equifax got a financial breach. Equifax was one of the top three credit reporting companies in the US (Daswani, 2021). This breach led to the exposing of personal data of around 147 million people. (Daswani, 2021). Which led to a settlement of \$700 million.

In July 2019, the news of the biggest financial breach came out. This breach has affected the financial sector. The breach was done by a former AWS worker named Paige A. Thompson who found a weakness in the system of capital one and used it to give access to personal information which was stored in the cloud . He gained access to the names, addresses, credit scores, social security numbers and the bank details of over one hundred thousand people. This breach led to the largest financial fine and Capital one paid over \$190 million to the customers who were affected by this breach. (Daswani, 2021)

The security violation caused the leak of several types of sensitive company :

- Names, addresses, phone numbers(140,000) (One, 2019)
- Bank account numbers (80,000) (One, 2019)

What was the Intention of the Breach?

- Analysts did not discover any financial reasons behind the attack because the stolen data was not sold.
- The internal attack seems motivated by personal struggles together with ideological reasons making Thompson appear a "rogue insider."

What were the Impact of the Breach?

- The public loss of trust toward their trusted financial institution .
- Capital One suffered from lawsuits which forced them to pay a \$190 million settlement according to Reuters (Daswani, 2021)

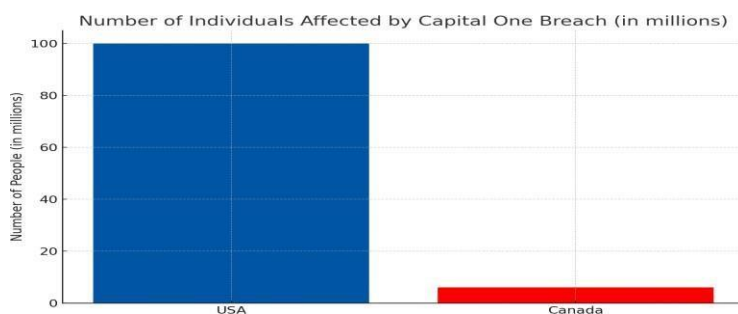


Figure 2 Number of individuals affected by the Capital One breach

2. Social Issues

Cybersecurity today shows us many social issues that include privacy violations for data breaches that exposes our personal information that can be used to steal money from us, our identity can be used for doing cybercrime without our information . (Azizi, 2023)

2.1 Background on Social Issues

The breach at Capital One led to many social concerns due to the exposure of millions of people's personal information that got exposed (Novaes Neto, 2020). In the era of digital world, privacy and data security are very important to keep safely and this creates trust between companies and their customers. When a data breach happens, this puts the customers' financial safety at risk and can damage their personal data. The breach at Capital One shows how easily cyberattacks can harm the society and can create Social Issues.

2.2 Identified Social Issues

Due to the breach of data in Capital One they didn't only face issues just in damage in finances they also faced Social Issues the lost the trust of the public .

1. Loss of Public Trust in Financial Institutions

The Capital One breach ruined public confidence in financial institutions while subscribers started doubting their interest in digital banking services (One, 2022). The extensive lack of trust spread financial sector instability because numerous users started looking for new banking institutions (Taddeo, 2017). Trust is foundational in banking, but such breaches erode it deeply. Such issues have affected customers, the digital finance industry and competing banks.

Stakeholders Affected : Customers, Digital finance industry and competing banks.

2. Emotional and Psychological Stress:

Due to the breached sensitive information disclosure victims experienced anxiety and slept poorly and went through emotional trauma which included sharing Social Security numbers (Console, 2024). Because of the fear of identity theft, it has used mental health impacts on users. Such issues have affected emotional as well as psychological stress to customers, mental health professionals, and consumer support organizations.

Stakeholders affected : Customers, Mental health professionals, and Consumer support organizations.

3. Increased Risk of Identity Theft and Financial Harm:

The Capital One breach made many survivors face instant dangers from identity theft and they needed to immediately freeze their credit and monitor their accounts. (McLean, 2019) People with lower incomes faced the sharpest consequences since their recovery required additional time and emotional strain (Richardson, 2025) this breach has triggered lack of financial insecurity . In this issue the stakeholders include victims, credit agencies and fraud preventive services

Stakeholders affected : Victims, Banks, Credit agencies, and Fraud prevention services.

4. Inequality in Digital Awareness and Protection

The security breach demonstrated widespread deficiencies regarding people's ability to use digital technology and technological services access. Tech-wizards promptly protected their information, yet elderly and less educated users found it difficult to comprehend both the incident and proper response strategies (Fier, 2019). Because of this gap in digital literacy, it has affected populations more exposed to. In such cases , inclusive digital education is much needed and preferred . It has affected elderly groups rural population and educational institutions as well as naïve citizens .

Stakeholders affected : Elderly groups , Rural population and educational institutions as well as naïve citizens .

5. Broader Societal Costs

Capital One suffered more than financial problems as a result of the breach incident. The investigation and support operations that government agencies conducted due to the breach led to higher public expenses. Public discussions about U.S. data privacy standards escalated after the breach as nation lacked a data protection framework comparable to the GDPR (Schroeder, 2020). This issue has not only caused financial loss to the company but also has the cost of government investigation . This issue has caused Data protection advocates , Taxpayers and Lawmakers

Stakeholders Affected : Data protection advocates , Taxpayers and Lawmakers .

3. Ethical Issues

Ethical Issues mean where a person or an organization must choose between right or wrong on the terms of moral principles. In the breach of Capital One the ethical issues came because the company failed to protect its own customers information for the breach.

3.1 Background on Ethical issues

The breach that occurs at Capital One causes exposed for more the 100 million customers personal details to be exposed. This incident shows us a lack of data management. We can see that Capital One as an organization fails to protect their customers information's which shows us a clear lack of ethical principles. Ethical failures , such as poor risk management and delayed telling the affected customers how negligent their ethical principles.

3.2 Identified Ethical Issues

1. Inadequate Data Protection

Capital One failed to give enough security measures to protect their customers sensitive data, this violates ethical principles of non-maleficence and justice. (digital privacy, 2023) . The past mirror cases like Equifax breach(2017) caused widespread damage which lead to poor risk management where customers regulatory bodies and employees were facing loss of trust , financial issues, and legal complications .

Comparison to Ethical theory:

Inadequate Data Protection – Deontology : The duty to protect personal data was not upheld.

Stakeholders affected : Customers, Regulators, Employees, and the public .

2. Delayed Disclosure

Capital One didn't inform the public about the breach and took over 4 months to inform them. Which led to people not getting information that was needed in time to protect them for possible harm. (Jones, 2022) Because of this delay it has increased the risk of identity theft for millions, in such cases timely communication is needed to decrease harm. The lack of response affected customers, regulators, security agencies, consumer rights organizations and the public.

Comparison to Ethical theory:

Delayed Disclosure – Utilitarianism : Failure to act promptly resulted in greater harm to the majority.

Stakeholders affected : Customers, regulators, security agencies, consumer rights organizations and the public.

3. Lack of Transparency

Capital One failed to provide a clear explanation for how serious the breach was. This led many customers to be confused and unsure how much of their data was leaked. (Wikipedia, 2025) Because of the lack of openness, it has reduced public trust. In such cases maintaining accountability is much needed to help make reliable decisions. The lack of Similar failures was seen in Facebook's data misuse such issues affected customers, advocacy groups and regulators.

Comparison to Ethical theory :

Lack of Transparency – Virtue Ethics: Lack of honesty and integrity in communication. Stakeholders affected : customers, advocacy groups and regulators.

Stakeholders affected : Customers, Advocacy Groups, Regulatory bodies.

4. Informed Consent Issues

Capital One didn't tell their customers how the data which got breach was collected and how it was being used. This raised concerns whether people can truly agree to share their information . (wikipedia, 2020) . The basic right has to be informed consent in data ethics without it the users will find hard to make educated choices about their privacy . A broader industry problem was affected where due to long unreadable policies the terms are unlikely to view . Such issues have affected customers, data protection authorities, and privacy advocacy groups.

Comparison to Ethical theory:

Informed Consent Issues – Contractarianism: Breach of the social contract by not obtaining proper consent.

Stakeholders affected : customers , advocacy groups, and data protection authorities .

5. Accountability and Responsibility

The breach raised questions to the customers about the accountability and responsibility of Capital One for safeguarding their customers data and the ethical implications of outsourcing data management to third parties. (Volkov, 2019) in such cases, Responsible data stewardship is much essential in today's digital economy. Because of the lack of accountability, it undermines the trust of the public. This breach has affected shareholders, regulatory bodies, and party vendors .

Comparison to Ethical theory :

Accountability and Responsibility - Ethics of Care : Neglecting the responsibility to care for and protect stakeholders

Stakeholders Affected : Shareholders, Regulatory bodies, and Party vendors .

3.3 Identified ethical issues to be compared with the general ethical theories.

Ethical Issue	Ethical Theory Violated	Explanation
Inadequate Data Protection	Deontological Ethics	Need for protecting personal information was not fulfilled.
Delayed Disclosure	Utilitarianism	Lack of action led to increased harm for the majority of people.
Lack of Transparency	Virtue Ethics	Absence of honesty and integrity in communication.
Informed Consent Issues	Contractarianism	Violation of the social contract because of the lack of legal consent.
Accountability and Responsibility	Ethics of Care	Failed to fulfill the responsibility to care for and protect stakeholders.

Figure 3 Ethical issues to be compared with the general ethical theories

4. Legal Issue

Legal issues mean that when a problem or a situation comes up and it involves interpretation, application, or violation of laws, this can come up when a person or a company or a government fails to follow any rules set by the legal system. For example, in 2019 Capital One data breach the company was faced with legal issues as they failed to secure sensitive data which violated the law of data protection . As a result, the company dealt with many court cases and investigations from official organizations which led to a huge fee of \$80 million from the OCC. As a result, companies found out that obeying data protection laws allows them to avoid problems and gain their customers' trust and because of the breach shareholders experienced a drop in the company's stock value. Offices faced with fines, costs to settle controversies and upset customers questioned how the company would perform overall.

4.1 Background on Legal Issues

In 2019 Capital One data breach exposed the personal information of over 100 million customers over US and Canada. This breach shows us multiple legal issues, like data protection obligations, responsibilities regarding third-party services, and the legal rights of affected customers. (One, 2022) The breach was investigated by U.S. federal court preliminarily and led to the settlement. This all-legal issues are related to both national and international data protection laws. (One, 2022)

4.2 Identified legal Issues

[The 5 legal issues are mentioned in appendix A](#)

4.3 Identified legal issues related with different laws, court cases, legal actions, fines and settlements and so on.

Identified legal issues related with different laws, court cases, legal actions, fines and settlements and so on. are mentioned in appendix B

5. Professional Issues

Those working in IT are required to act with integrity and take responsibility for their work. What they do impacts users, companies, and society greatly. In this case, a professional means anybody working in technology, such as software engineering, cybersecurity, or data management, and adhering to ethical guidelines set out by ACM, IEEE, or BCS. In the case of Capital One we can see the lack of proactive measures to protect customer data. This raised the question of whether these ethical standards were fully upheld. Those in charge of security were requested to point out any possible threat to cloud storage and amend protect the data. We see from this incident just how important it is for IT experts to protect systems from such dangers. They should always focus on keeping stakeholders safe and trusting, ensuring that new technology does not harm their ethics.

5.1 Background on Professional Issues

When people or organizations drift away from proper rules in managing data, talking, or delegating tasks, this can lead to professional issues. By looking into the Capital One data breach, it was clear that there were flaws in assessing risk, managing the cloud, and regulating data, all showing errors in how things are managed inside. (The OCC Fines Capital One Bank for 2019 Cybersecurity Breach , 2019)

5.2 Identified Professional Issues

[The 5 Professional Issues are mentioned in appendix C](#)

5.3 Identified Professional issues to be compared with the Professional Code of Conduct

Issue	Explanation	Related Professional Codes of Conduct(ACM, SE, IEEE, BCS etc.)
1. Inadequate Risk Assessment	The company failed to review the risks properly and ended up having important information leaked when data was stored in the cloud.	ACM Code 2.5: Conduct full and focused reviews of systems based on risks that could occur.
2. Misconfiguration of Firewall	A firewall mistake allowed more than 100 million records to be accessible to individuals.	The concepts of public health, privacy, security and well-being when conducting your work.
3. Delayed Incident Disclosure	It took the company four months to release the information, customers did not have a chance to guard themselves against breaches.	IEEE Code #5: If you make a mistake, try to fix it and remain open when discussing matters in engineering.
4. Lack of Competent Oversight	People overseeing the migration of data ignoring cybersecurity, resulting in issues that could have been avoided.	Code 1.2: Remember that overlooking security could be devastating to your work.
5. Poor Communication Practices	Capital One did not provide clear and complete information which left others misinformed.	For SE Code, ever make sure you speak the truth and don't attempt to deceive anyone publicly.

6. Conclusion

The Capital One data breach of 2019 is a prime example of important issues in cybersecurity, ethics, the law, and responsible behavior. The findings underline how major companies with resources can still expose customer data because of a lack of proper safety protocols. A compromise of hard-to-mask personal and financial details came about when a misconfigured firewall and poor risk assessment were not managed, leading to the loss of over 100 million records. (Office of the Comptroller of the Currency Issues Proposed True Lender Rule, 2020)

The report looked at the breach from several sides: social, ethical, legal, and professional. The breach lowered people's confidence in financial systems, stressed individuals emotionally, and raised the chance of personal information being stolen. Many watched unimpressed at how Capital One failed to be clear, get proper consent, and maintain accountability for their data handling. The company was liable to face tough fines, close monitoring by regulators, and lawsuits due to data protection violations, as the GDPR or the Fair Credit Reporting Act became more important. It was seen in this incident that risk management and overseeing audits were not properly followed as per the standards of ACM and IEEE.

So, protecting data demands the use of technology, proper ethics, and following laws. It means that professionals must care for and responsibly manage private data when they receive it. Once the breach happened, Capital One put security measures in place and collaborated with authorities, but experts view it as a lesson on how fast cyber-crimes can damage any company.

6.2 Personal Reflection

This work has allowed me to better see how complex cybersecurity can be when seen through different ethical and legal aspects. Recognizing the damage caused by the Capital One data breach revealed to me the serious impact that one error can have on people and on the reputation of corporations. I have found that data breaches are more than just errors in security; they are also incidents where the company fails to do what is ethically right. I learned from this that we need to be proactive in leadership and to take responsibility for everything in the digital era.

6.3 Recommendations

Use an Ethics Decision-Making Approach In order to stand clear of more scandals like Capital One faced, following Reynolds 6-step ethical decision-making process is recommended.

1. Identify the Ethical Issue

Due to technological problems and sudden announcements, capital one did not properly secure clients' information.

2. Gather Relevant Facts

A security breach occurred because the firewall was set up incorrectly on AWS. They are in violation of PIPEDA/GDPR because they shared the information after the deadline. This means that the data of over 100 million people was visible to others.

3. Evaluate Alternatives Using Ethical Theories

With AI for detection, learning about ethics and adopting Privacy by Design, we are able to avoid problems for everyone, obey rules and regulations and make our actions transparent (respectively).

4. Make a Decision

By including Security of Law and Disclosure Protections, it is important to promote ethical and responsible behavior.

5. Implement the Decision

Employ ethical hackers to check and secure your systems (in the near future) and persuade the government to include GDPR-type policies (for the future).

6.4 References

Anon., 2019. *capitalone*. [Online]

Available at: <https://www.capitalone.com/about/corporate-information/our-company/> [Accessed 17 may 2025].

Azizi, N. a. H. O., 2023. *igi-global*. [Online]

Available at: <https://www.igi-global.com/chapter/cybersecurity-issues-and-challenges/314073> [Accessed 17 may 2025].

cisoplatfrom, 2025. *cisoplatfrom*. [Online]

Available at: <https://www.cisoplatfrom.com/profiles/blogs/executive-summary-of-capital-one-data-breach-analyzing-compliance#:~:text=Capital%20One's%20breach%20affected%20over,applied%20to%20enhance%20security%20posture.>

[Accessed 17 May 2025].

Console, 2024. *databreachclassaction*. [Online]

Available at: <https://databreachclassaction.io/blog/the-psychological-impact-of-data-breaches-on-victims>

[Accessed 17 MAY 2025].

Daswani, N. E. M., 2021. *The Capital One Breach*. In: *Big Breaches*. , Berkeley: s.n.

Daswani, N. E. M., 2021. *The Equifax Breach*, Berkeley: Apress, Berkeley, CA.

digital privacy, 2023. *digitalprivacy*. [Online]

Available at: <https://>

[Accessed 17 May 2025].

Fier, J., 2019. *Lessons from the Capital One Breach on Cloud Security*, s.l.: s.n.

Folorunso, A. M. V. W. I. a. S. B., 2024. *The impact of ISO security standards on enhancing cybersecurity posture in organizations.*, s.l.: World Journal of Advanced Research and Reviews. Ghelani, D., 2022. *authorea.com*. [Online]

Available at: <https://www.authorea.com/doi/full/10.22541/au.166385207.73483369> [Accessed 17 may 2025].

Greenstein, S., 2019. *The Aftermath of the Dyn DDOS Attack*, s.l.: s.n. Jones, T., 2022. *medium.com*. [Online]

Available at: <https://medium.com/nerd-for-tech/capital-one-data-breach-2019-f85a259eaa60> [Accessed 17 May 2025].

Kao, S. -C. H. a. D. -Y., 2018. *The static analysis of WannaCry ransomware.*, Chuncheon: Communication Technology.

McLean, R., 2019. *A hacker gained access to 100 million Capital One credit card applications and accounts*, s.l.: Cnn .

Novaes Neto, N. M. S. M. G. d. P. A. a. M. B. N., 2020. *A case study of the capital one data breach*, s.l.: s.n.

Office of the Comptroller of the Currency Issues Proposed True Lender Rule, 2020. *OCC.gov*. [Online]

Available at: <https://www.occ.gov/news-issuances/news-releases/2020/nr-occ-2020-97.html>. [Accessed 17 may 2025].

Olivér Gulyás, G. K., 2023. *Impact of cyber-attacks on the financial institutions*, s.l.: Volume

219.

One, C., 2019. *Capital One*. [Online]

Available at: <https://www.capitalone.com/digital/facts2019/> [Accessed 17 May 2025].

One, C., 2022. *Capital One*. [Online]

Available at: <https://www.capitalone.com/digital/facts2019/> [Accessed 17 MAY 2025].

Richardson, A., 2025. *dailyhodl*. [Online]

Available at: <https://dailyhodl.com/2025/04/04/thousands-of-americans-personal-information-exposed-by-banking-giant-capital-one-alleges-new-class-action-lawsuit/>

[Accessed 17 May 2025].

Schopen, i. R., 2025. *courtlistener*. [Online]

Available at: <https://www.courtlistener.com/docket/16967199/in-re-capital-one-consumer-data-security-breach-litigation/>.

[Accessed 17 May 2025].

Schopen, N. R., 2025. *courtlistener*. [Online]

Available at: <https://www.courtlistener.com/docket/16967199/in-re-capital-one-consumer-data-security-breach-litigation/>

[Accessed 17 May 2025].

Schroeder, P., 2020. *Capital One to pay \$80 million fine after data breach*, Reuters, s.l.: s.n. Solove, 2006. s.l.: s.n.

Taddeo, 2017. s.l.: s.n.

Team T., 2023. <https://lexchart.com/org-charts/capital-one-ownership-structure-2024/#:~:text=Richard%20Fairbank%20is%20the%20founder,24.85%25%20stake%20in%20the%20company..> [Online]

Available at: <https://lexchart.com/org-charts/capital-one-ownership-structure-2024/#:~:text=Richard%20Fairbank%20is%20the%20founder,24.85%25%20stake%20in%20the%20company.>

[Accessed 17 May 2025].

The OCC Fines Capital One Bank for 2019 Cybersecurity Breach , 2019 . *Weave*. [Online] Available at: <https://weaver.com/resources/occ-fines-capital-one-bank-2019-cybersecurity-breach/>

[Accessed 17 May 2025].

Volkov, M., 2019 . *blog.volkovlaw.com*. [Online]

Available at: <https://blog.volkovlaw.com/2019/09/lessons-learned-from-the-capital-one-data-breach-part-i-of-iii/>

[Accessed 17 May 2025].

Volkov, M., 2019. *blog.volkovlaw*. [Online]

Available at: <https://blog.volkovlaw.com/2019/09/lessons-learned-from-the-capital-one-data-breach-part-i-of-iii/>

[Accessed 17 May 2025].

wikipedia, 2020. *wikipedia.org*. [Online]

Available at: https://en.wikipedia.org/wiki/2020_Twitter_account_hijacking [Accessed 17 May 2025].

wikipedia, 2025 . *wikipedia.org*. [Online]

Available at: https://en.wikipedia.org/wiki/Capital_One#2019_data_breach [Accessed 17 May 2025].

Zafar, M., 2021. <https://link.springer.com/article/10.1007/s11235-020-00733-2>. [Online] Available at: <https://link.springer.com/article/10.1007/s11235-020->

7. Appendix

7.1. The 5 legal issues are mentioned in appendix A

1. Violation of Data Protection Laws

The company did not meet safety standards for protecting private customer data and many laws were broken. breach laws, as well as Canada's laws. As a result, hackers can now access and read people's personal details. (cisoplatfrom, 2025) The main people affected where customers since their personal information was taken, making them face fraud and identity theft .The company needs to be investigated, the regulators began reviewing their guidelines for handling user data.

Stakeholders Affected : Customers, regulators , Legal and Compliance team .

2. Negligence in Security Measures

Because of an incorrectly set security system on Amazon Web Services (AWS), someone was able to hack the company, showing that cybersecurity was not well executed. There are legal issues about Capital One's efforts to protect customer data, because their safeguards were insufficient.. (Schopen, 2025) Because of the security breach, those responsible for the IT and cybersecurity groups were scorned by the public and dealt with criticism from the law. Due to low stock prices and people losing trust in the company, shareholders faced losses.

Stakeholders affected : It and Cybercity teams, Customers and Investors.

3. Third-Party Liability and Outsourcing Risks

Having their critical data placed in the cloud using AWS has led to discussions on corporate responsibility and who is to blame. It highlighted the need to have strict agreements in place and watch over the actions of third-party service providers. (Volkov, 2019) Such security breaches lowered the reputation of companies like AWS, forcing them to consider strengthening their security rules. Firms relying on cloud storage also faced problems, since people's confidence in such services was shaken.

Stakeholders Affected : Legal teams , Compliance teams and public .

4. Failure to Notify Affected Customers Promptly

The company took several months to send customers notifications, ignoring data breach notification requirements for helping clients. The decision to hide the information stopped customers from acting in time to secure themselves. (Office of the Comptroller of the Currency Issues Proposed True Lender Rule, 2020) People were not informed and could not react quickly, leading to additional issues for their emotions. Such companies faced criticism for not making it possible to obtain information more quickly.

Stakeholders Affected : Customers , Regulatory authorities and Legal teams .

5. Class-Action Lawsuits and Consumer Rights

Groups of customers in the lawsuit accused Capital One of negligence and claimed to have suffered emotional distress. They indicate that consumers are entitled to claim damages if their data protection is not observed. (Schopen, 2025) It was costly for Capital One to oversee lawsuits, settlements and penalties from regulators. Damage to the company's brand and reputation caused customers to lose trust in the business for a long period.

Stakeholders Affected : Customers, Judicial system and regulatory Bodies .

[Return to page no 17](#)

7.2. Identified legal issues related with different laws, court cases, legal actions, fines and settlements and so on. are mentioned in appendix B

Legal Issue	Explanation	Related Laws / Regulations	Court Cases / Legal Actions	Fines / Settlements
1. Violation of Data Protection Laws	Weak safeguarding of personal data and violations with data security standards.	U.S. State Data Breach Notification Laws, PIPEDA (Canada)	investigations by Canadian security authorities and state attorneys general.	investigation by regulators; FTC monitoring. (cisoplatform, 2025)
2. Negligence in Security Measures	Incorrect security systems led to illegal access of data hosted on AWS.	General negligence under civil law	Lawsuit for Capital One Consumer Data Security Breach (class-action)	Part of \$190M class-action settlement (One, 2022)
3. Third- Party Liability (AWS Hosting)	Legal issues about liability for cloud breaches were brought up by the use of third parties (Amazon Web Services).	Contract law, SLA obligations, cybersecurity outsourcing laws	Capital One was held accountable for vendor misconfigurations, although there was no direct legal action taken against AWS.	Legal debates on cloud responsibility; no AWS penalty noted (Volkov, 2019)
4. Delay in Customer Notification	Users were not notified for more than four months, which is against breach notification regulations.	U.S. State Laws, NY SHIELD Act, GDPR Article 33 (if applicable)	Regulatory scrutiny for delayed disclosure	OCC fined \$80 million for “failure to establish effective risk management.” (Volkov, 2019)

5. Consumer Class-Action Lawsuits	lawsuits filed by impacted users who claimed a lack of protection, grief, and the potential of identity theft.	Consumer protection laws,	Capital One Consumer Data Security Breach Litigation (Eastern District of Virginia)	\$190 million class-action settlement paid in 2022 (One, 2022)
-----------------------------------	--	---------------------------	---	--

Figure 4 legal issues related to different laws

[Return to page number 18](#)

7.3. Identified Professional Issues are mentioned in appendix C

1. Inadequate Risk Assessment

Capital One made the mistake of not assessing risks in detail before uploading customers' data to the cloud. This situation resulted in large security holes that gave unauthorized individuals access to millions of customers' data. When we consider the ACM Code, this means that it goes against 2.5, which clearly advises checking computers and their risks in detail. Customers suffered the most because their data was disclosed because of poor planning. CapitalOne was also criticized because its cybersecurity experts did not discover these risks earlier.

Stakeholders Affected : Customers , Chief Information Security Officer (CISO), investors , Regulatory bodies and Capital One's Board of Directors.

2. Misconfiguration of Firewall

The firewall on the Capitol One system was not properly set up by chance, allowing a previous employee to break in and take over 100 million records. Because of this incident, it appeared that the professionals involved were not following BCS rules, since the Code requires professionals to be conscious of and respect the privacy, security and well-being of everyone. Not only were customers affected, but IT staff and engineers were, too, because their issues destroyed confidence in the company and resulted in responsibility for their actions. As a result, the organization faced strict regulation and suffered financial penalties.

Stakeholders Affected : Capital One Network Security Engineers , Customers , regulatory bodies and Capital One Compliance Department .

3. Delayed Incident Disclosure

It took capital One over four months to inform customers that they had been involved in the breach. Since the delay resulted in no immediate action, many victims had to act only after the incident. This situation is not in line with the IEEE Code of Ethics #5 which requires us to correct any mistakes and clearly explain any failures. Many customers were made unaware that their financial lives could be seriously affected. Moreover, if businesses do not report their data breaches properly, it can lead to problems and investigations by regulators and the courts.

Stakeholders Affected : Customers , Employees , and Legal authorities .

4. Lack of Competent Oversight

People responsible for managing data and security were not attentive and did not understand the potential risks involved. What happened here violates principle 1.2 of the ACM Code of Ethics as it urges IT professionals to “not harm.” These gaps turned out to be very damaging due to poor

security. It affected all stakeholders including executives, employees and customers. People dealing with the company lost their personal information and were struggling to trust it, while its employees went through a review process and new organizational changes. They were put under pressure by both rules and investors.

Stakeholders Affected : Customers , Shareholders , Legal and Compliance Department .

5. Poor Communication Practices

Information provided by Capital One in the aftermath was not detailed enough and lacked transparency. Little information about what happened confused the public, the press and stakeholders. The engineers violated the Software Engineering Code of Ethics because it requires being honest and truthful in conversations, especially those that happen in public. Those concerned were unsure about the next thing they could do to keep their information safe. Missing certain details from the media and the firm meant investors were not satisfied and acted negatively.

Stakeholders Affected : Customers , Legal & Regulatory Compliance Team at Capital One and Consumer Advocacy Groups

[Return to Page number 19](#)